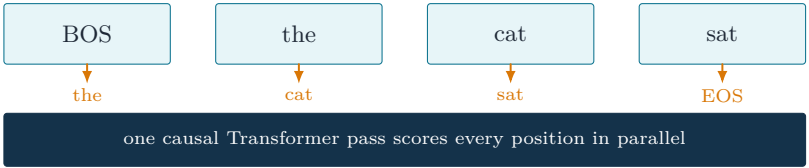
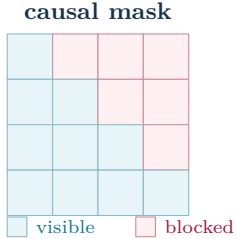
$$\log p_\theta(x) = \sum_{t=1}^{T} \log p_\theta(x_t \mid x_{<t})$$

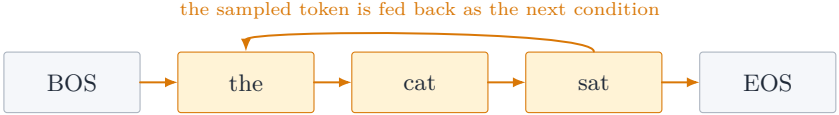
TRAIN
teacher forcing
real prefixes



loss $= \sum_t -\log p_\theta(x_t \mid x_{<t})$ — a single sum over all t



DECODE
ancestral
sampling



each token is drawn from $p_\theta(x_t \mid x_{<t})$ and appended to the prefix
so step t conditions on the model's own output, not on ground truth

exposure bias
train: real prefixes
decode: the model's own prefixes